Sample Research Note: Adaptive Patch Sampling for Self-Supervised Learning

Abstract

This synthetic note exists so a first-time user can run the ingest workflow end to end without supplying their own document. Replace it with your own PDF.

1. Motivation

Self-supervised models learn from patches cropped out of unlabeled images. Most methods sample patches uniformly at random. We hypothesize that sampling patches in proportion to local gradient magnitude yields more informative training signal under a fixed compute budget.

2. Method

At each step we estimate a saliency map from the running loss gradient and draw patch centers from a distribution proportional to that map. The sampler is refreshed every epoch to track the model's changing uncertainty.

3. Expected Result

We expect higher linear-probe accuracy at equal numbers of seen patches, especially early in training when the model benefits most from hard regions.

4. Limitations

Dynamic sampling adds bookkeeping overhead and may destabilize training if the saliency map is refreshed too aggressively.